V. Complete the analogy.

1. Knot in a thread : _____ friction :: Ball bearing : _____ friction
2. Downward force : Weight :: Upward force offered by liquid : _____

VI. Numerical Problem.

1. A stone weighs 500 N. Calculate the pressure exerted by it, if it makes contact with a surface of area 25 cm^2.

VII. Consider the statements given below and choose the correct option.

1. **Assertion:** Sharp knives are used to cut the vegetables.
   Reason: Sharp edges exert more pressure.
2. **Assertion:** Broad straps are used in bags.
   Reason: Broad straps last for long.
3. **Assertion:** Water strider slides easily on the surface of water.
   Reason: Water strider experiences less buoyant force.
   a. Both assertion and reason are true and reason is the correct explanation of assertion.
   b. Both assertion and reason are true, but reason is not the correct explanation of assertion.
   c. Assertion is true, but reason is false.
   d. Both assertion and reason are false.

VIII. Answer very briefly.

1. Give two examples to verify that a force changes the shape of a body.
2. Give two examples to verify that a force tends to change the static condition of a body.
3. How do you feel when you touch a nail immediately after it is hammered into a wooden plank? Why?
4. How does the friction arise between the surfaces of two bodies in relative motion?
5. Name two instruments which help to measure the pressure of a fluid.
6. Define one atmosphere.
7. Why are heavy bags provided with broad straps?
8. How does surface tension help a plant?
9. Which has greater viscosity, oil or honey? Why?

IX. Answer briefly.

1. Define friction. Give two examples of the utility of friction in day to day life.
2. Mention any three ways of minimising friction.
3. State Pascal's law and mention its applications.
4. Why is a ball bearing used in a cycle hub?

X. Answer in detail.

1. Friction is a necessary evil - Explain.
2. Give the different types of friction and explain each with an example.
3. Describe an experiment to prove that friction depends on the nature of a surface.
4. Explain how friction can be minimised.
5. Describe an experiment to prove that the pressure in a liquid increases with depth.

XI. Higher Order Thinking Questions.

1. Why is it not advisable to use a fountain pen while travelling in an aeroplane?
2. Is there any possibility of making a special device to measure the magnitude of friction directly?
3. Vidhya feels that mercury is costly. So, instead of mercury she wants to use water as a barometric liquid. Explain the difficulty of constructing a water barometer.

XII. Project Work.

Observe the devices, gadgets or things around you. List out the types of friction involved in each device. How would you minimise the friction? Record your observations and discuss your results with your classmates.

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